

VOYAGE ANALYTICS: MLOPS INFRASTRUCTURE & DEPLOYMENT DOCUMENTATION

Author: Sanidhya Shekhar

Project: [Voyage Analytics - Integrating MLOps in Travel](#)

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CHAPTER 1: PROJECT OVERVIEW & ARCHITECTURE

Voyage Analytics is an end-to-end MLOps platform engineered to bring production-ready machine learning solutions to the travel and tourism sector. This framework integrates three core intelligent engines:

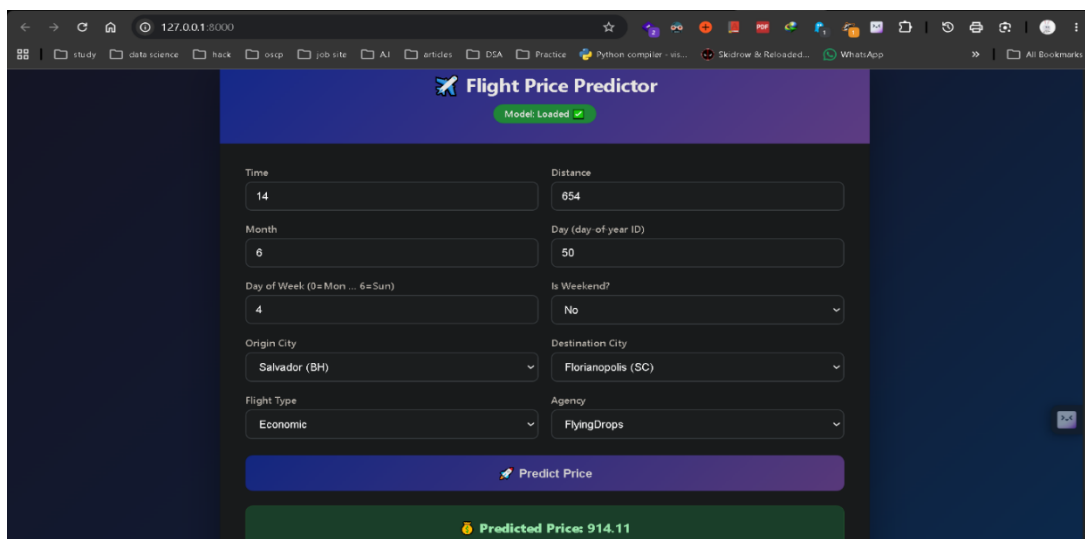
Flight Price Prediction: Fully automated via a robust MLOps pipeline covering automated scheduling, continuous integration, containerized scaling, and experiment tracking.

1. REGRESSION MODEL DEVELOPMENT (FLIGHT PRICE PREDICTION)

- **Objective:** Built a robust regression model using the flights.csv dataset to accurately predict flight prices. The pipeline incorporates strict feature engineering, selection, training validation, and evaluation to ensure maximum model reliability.

2. REST API FOR REGRESSION MODEL (FLASK)

- **Objective:** Developed a real-time prediction serving layer using a Flask REST API (flight_price_api.py). It exposes web endpoints (such as /predict) to dynamically serve the serialized model artifact.
- **System View:**



The screenshot displays a web application titled "Flight Price Predictor" with a status indicator "Model: Loaded". The interface features a dark-themed layout with input fields for various flight parameters. The inputs are as follows:

Parameter	Value
Time	14
Distance	854
Month	6
Day (day of year ID)	50
Day of Week (0=Mon ... 6=Sun)	4
Is Weekend?	No
Origin City	Salvador (BH)
Destination City	Florianopolis (SC)
Flight Type	Economic
Agency	FlyingDrops

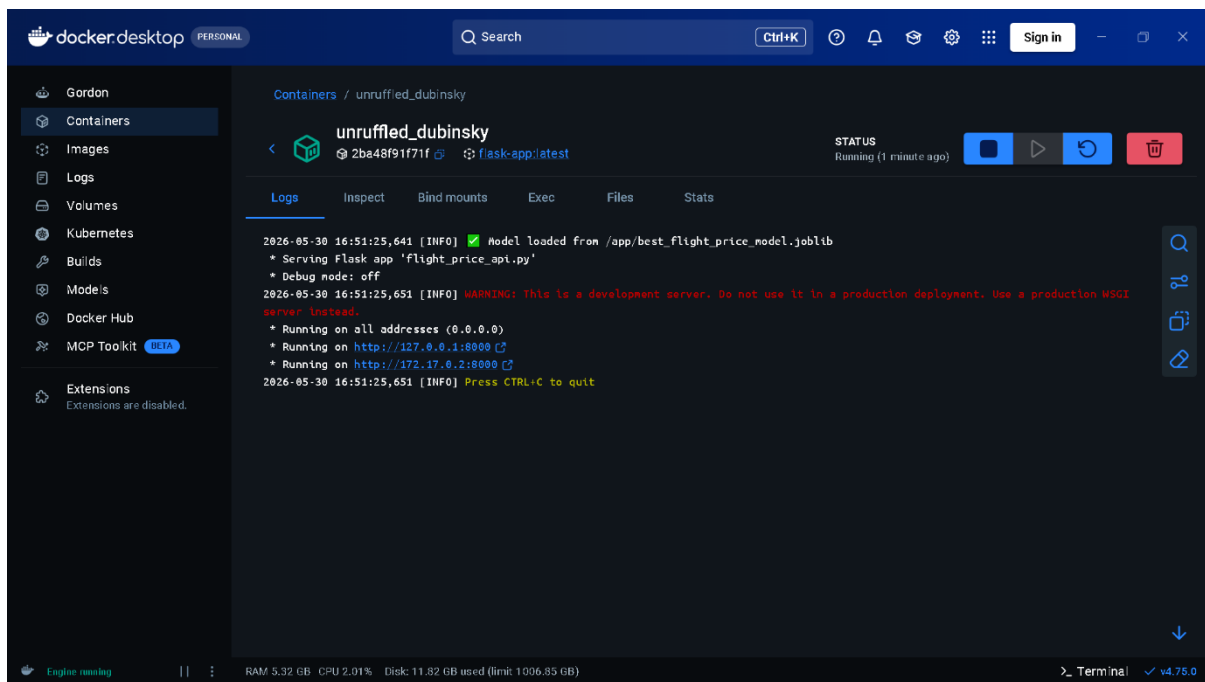
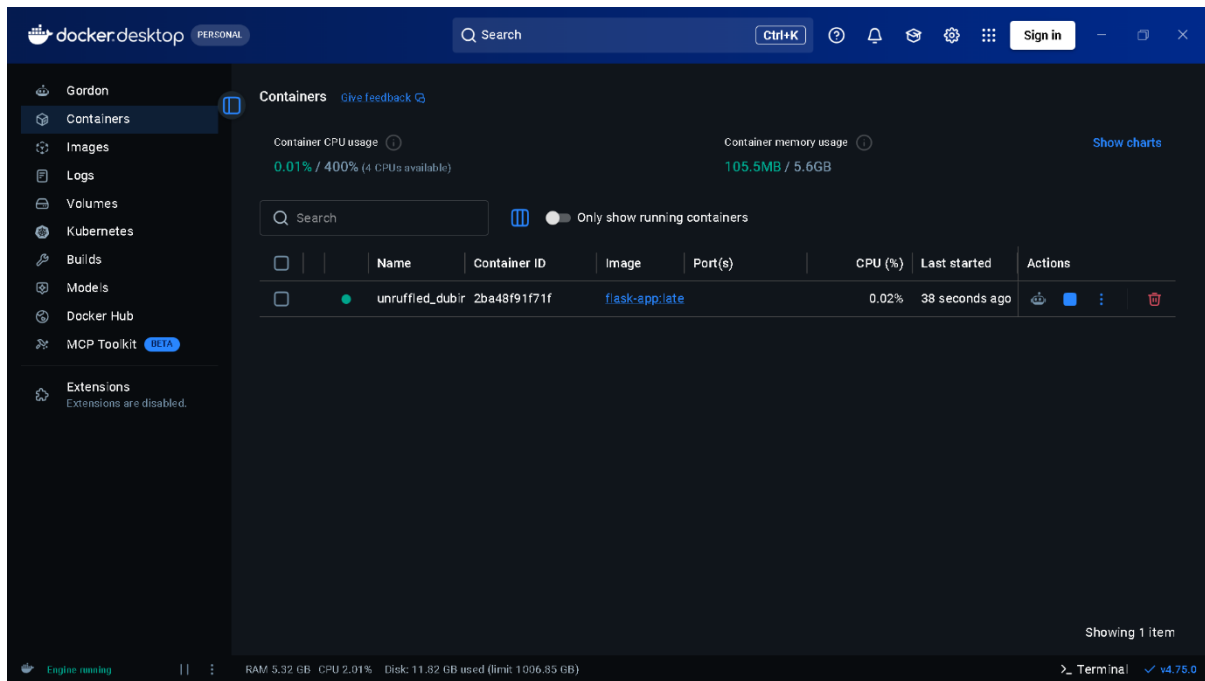
A "Predict Price" button is located below the input fields. At the bottom of the form, the "Predicted Price" is displayed as 914.11.

Caption: Active Flask REST API endpoint returning automated, real-time flight price prediction JSON payloads.

3. CONTAINERIZATION (DOCKER)

- **Objective:** Packaged the flight price prediction application and its dependencies into a Docker container image to ensure environmental portability, isolated execution environments, and flawless deployment across infrastructure targets.

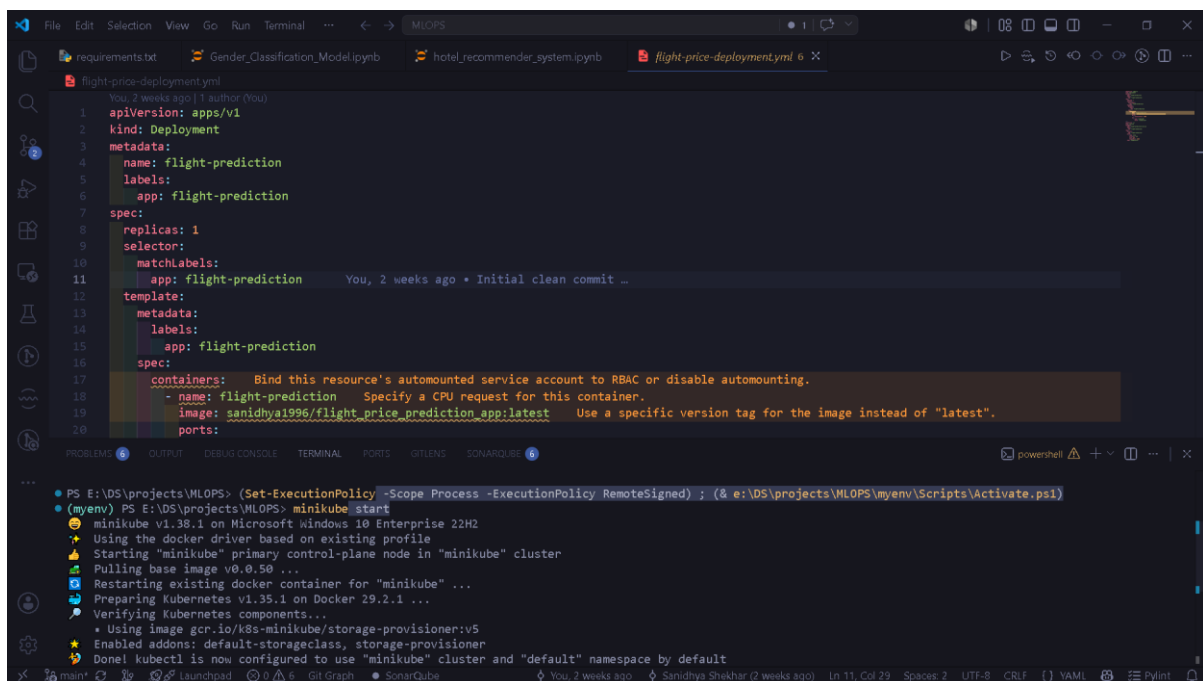
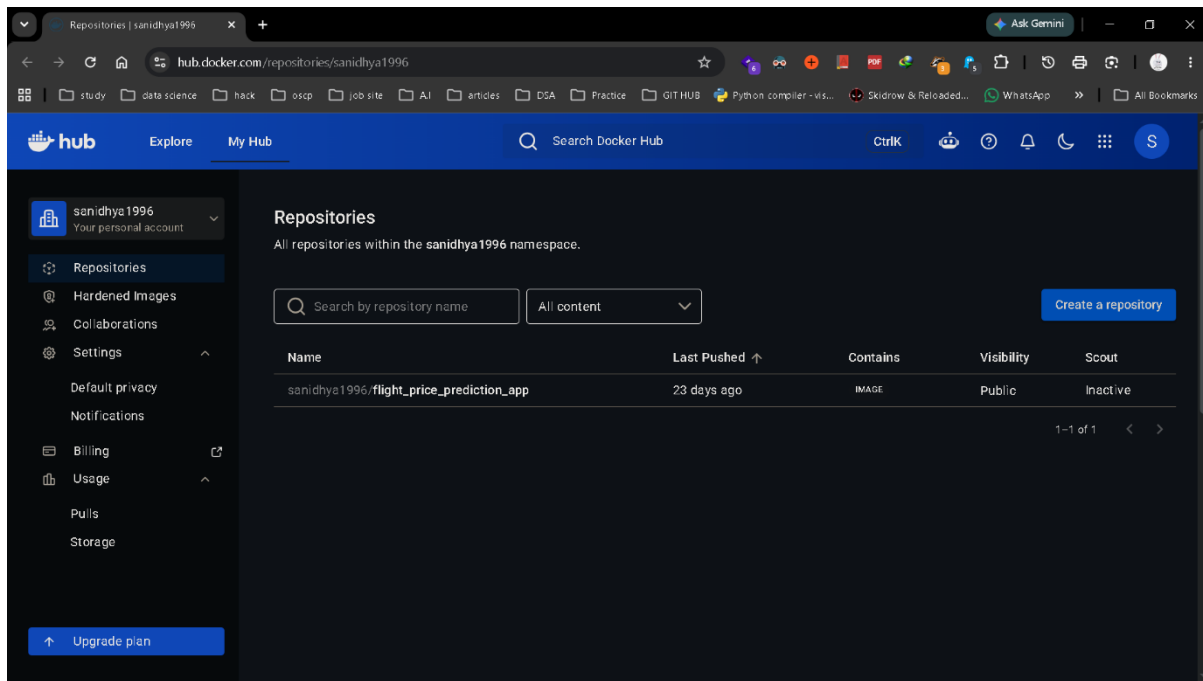
System View:



Caption: Dockerized application engine illustrating isolated, lightweight container states for the model environment.

4. KUBERNETES FOR SCALABILITY

- **Objective:** Orchestrated the containerized flight price API inside a Kubernetes (K8s) cluster using custom manifests (flight-price-deployment.yaml). This manages auto-scaling, load balancing, and ensures high-availability under varying traffic volumes.
- **System View:**



```
20 ports:
PROBLEMS 6 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SONARQUBE 6
...
(myenv) PS E:\DS\projects\MLOPS> kubectl get all
NAME                                READY    STATUS    RESTARTS      AGE
pod/flight-prediction-df9575848-d8smn 1/1      Running   1 (112s ago)   21d

NAME                                TYPE          CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
service/flight-prediction-service    LoadBalancer 10.108.214.189 <pending>      80:31859/TCP     21d
service/kubernetes                   ClusterIP      10.96.0.1     <none>         443/TCP          22d

NAME                                READY    UP-TO-DATE    AVAILABLE    AGE
deployment.apps/flight-prediction    1/1      1             1            21d

NAME                                DESIRED    CURRENT    READY    AGE
replicaset.apps/flight-prediction-df9575848 1          1          1        21d

Kubernetes Info
(myenv) PS E:\DS\projects\MLOPS> kubectl cluster-info
Kubernetes control plane is running at https://127.0.0.1:61192
CoreDNS is running at https://127.0.0.1:61192/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
(myenv) PS E:\DS\projects\MLOPS>
```

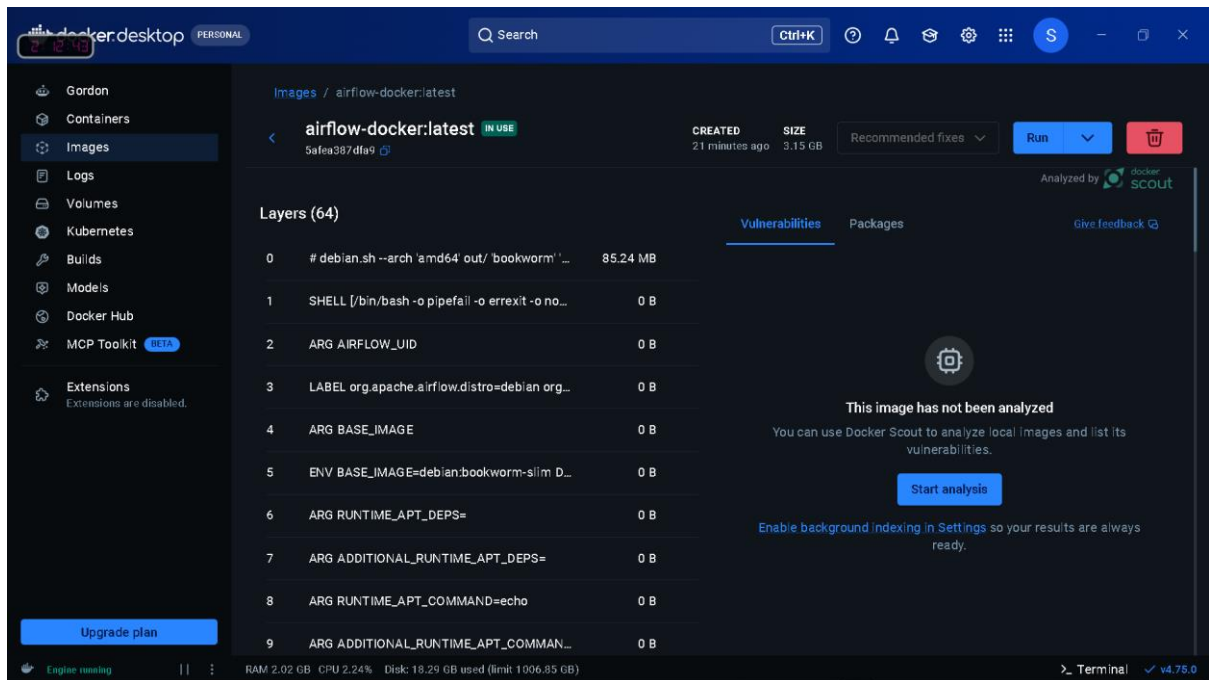
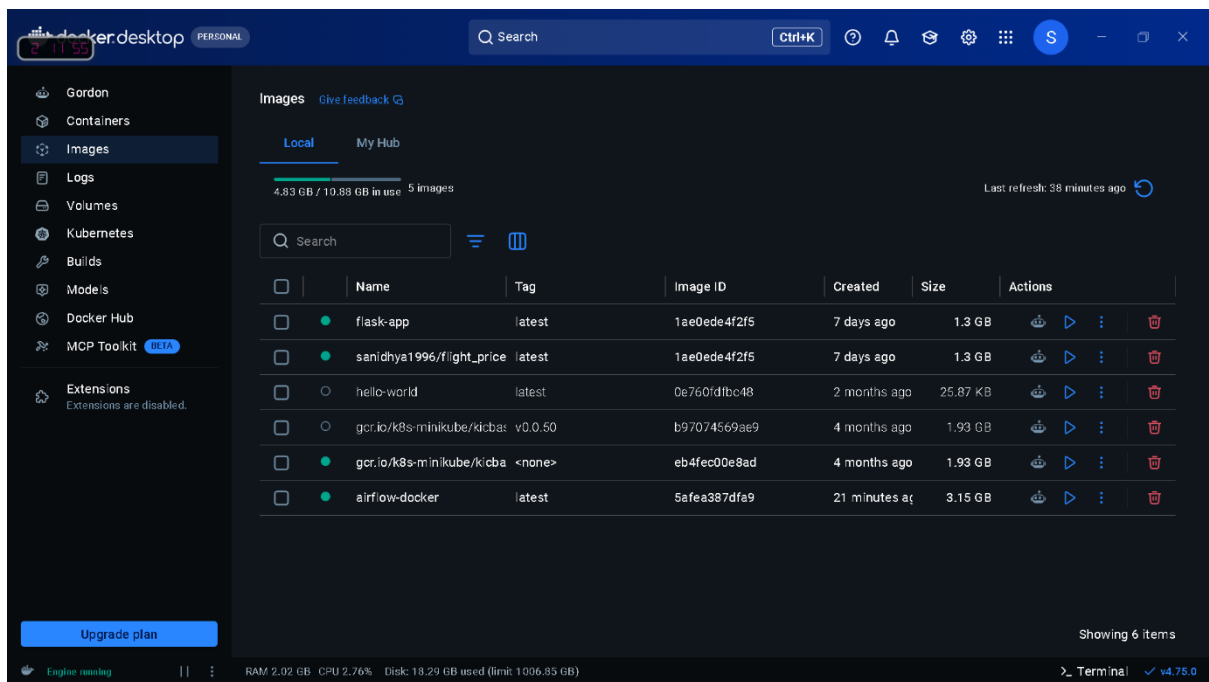
```
(myenv) PS E:\DS\projects\MLOPS> kubectl apply -f flight-price-deployment.yml
deployment.apps/flight-prediction unchanged
service/flight-prediction-service unchanged
(myenv) PS E:\DS\projects\MLOPS> kubectl get deployments
NAME                                READY    UP-TO-DATE    AVAILABLE    AGE
flight-prediction                  1/1      1             1            21d
(myenv) PS E:\DS\projects\MLOPS>
```

```
PROBLEMS 6 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SONARQUBE 6
...
PS E:\DS\projects\MLOPS> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& e:\DS\projects\MLOPS\myenv\Scripts\Activate.ps1)
(myenv) PS E:\DS\projects\MLOPS> minikube tunnel
Tunnel successfully started

NOTE: Please do not close this terminal as this process must stay alive for the tunnel to be accessible ...

! Access to ports below 1024 may fail on Windows with OpenSSH clients older than v8.1. For more information, see: https://minikube.sigs.k8s.io/docs/handbook/accessing/#access-to-ports-below-1024-on-windows-requires-root-permission
Starting tunnel for service flight-prediction-service.
```

```
(myenv) PS E:\DS\projects\MLOPS> kubectl get services
NAME                                TYPE          CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
flight-prediction-service          LoadBalancer 10.108.214.189 127.0.0.1      80:31859/TCP     21d
kubernetes                         ClusterIP      10.96.0.1     <none>         443/TCP          22d
(myenv) PS E:\DS\projects\MLOPS>
```

airflow

localhost:8080/dags

Search Dags

All Failed Queued Running Success Required Actions All Active Paused Filter by tag

All Favorite Unfavorite

1 Dag

Sort by Display Name (A-Z)

flight_price_prediction_dag

Schedule Latest Run Next Run

00*** 2026-06-08 13:20:35 2026-06-09 05:30:00

flight_price_prediction_dag - Air

localhost:8080/dags/flight_price_prediction_dag

Search Dags Trigger

flight price prediction_dag

Schedule Latest Run Next Run Max Active Runs Owner Tags Latest Dag Version

00*** 2026-06-08 13:20:35 2026-06-09 05:30:00 0 of 16 Admin v1

Overview Runs Tasks Calendar Backfills Audit Log Code Details

Last 24 Hours 2026-06-07 13:29:53 - 2026-06-08 13:29:53

0 Failed Tasks 0 Failed Runs

Last 2 Dag Runs

Queued Duration Run Duration

00:05:00 00:03:20 00:01:40 00:03:37 00:00:00

05:30:00 13:20:35 Run After

Airflow interface showing the DAG Run for `flight_price_prediction_dag` (manual_2026-06-08T07:50:35.625723+00:00). The DAG is marked as **Success**.

DAG Run Summary:

Logical Date	Run Type	Start Date	End Date	Duration	Triggering User	Dag Version(s)
2026-06-08 13:20:24	Manual	2026-06-08 13:20:36	2026-06-08 13:25:06	00:04:30	admin	v1

Task Instances:

Task ID	Map Index	State	Start Date	Try Number	Operator	Duration
random_forest_task		Success	2026-06-08 13:23:09	1	PythonOperator	00:01:...
transform_data_task		Success	2026-06-08 13:20:58	1	PythonOperator	...
load_data_task		Success	2026-06-08 13:20:43	1	PythonOperator	00:00:...

Airflow interface showing the Task Run for `load_data_task` (manual_2026-06-08T07:50:35.625723+00:00). The task is marked as **Success**.

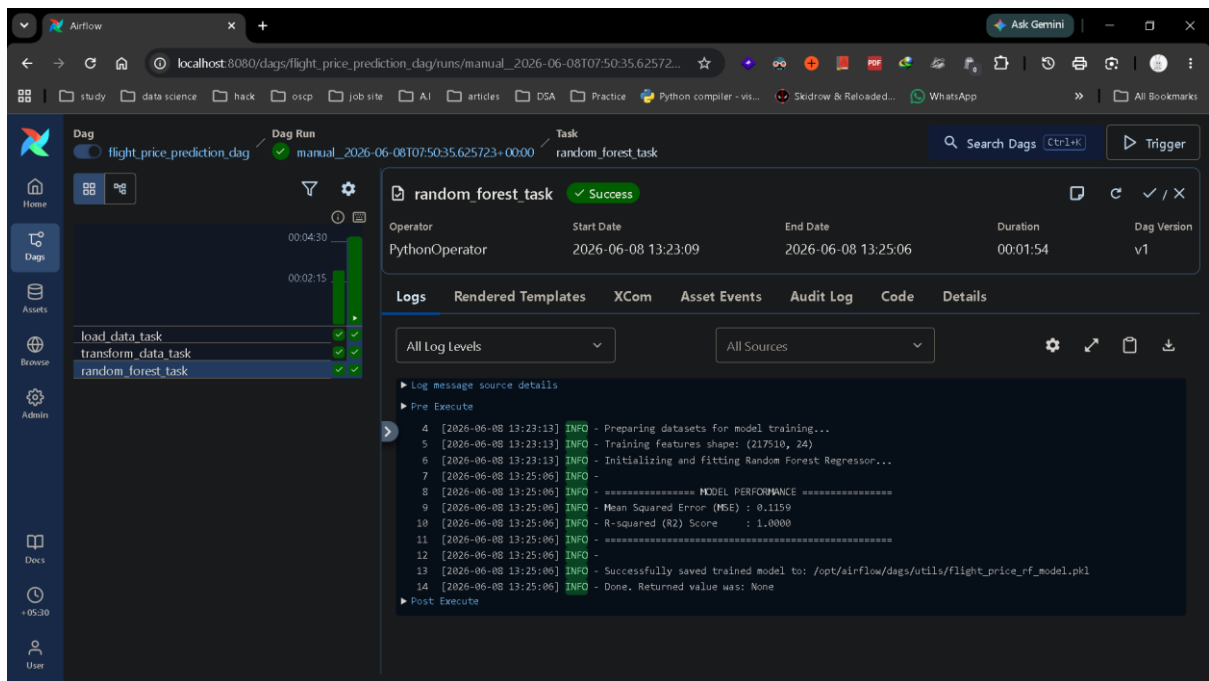
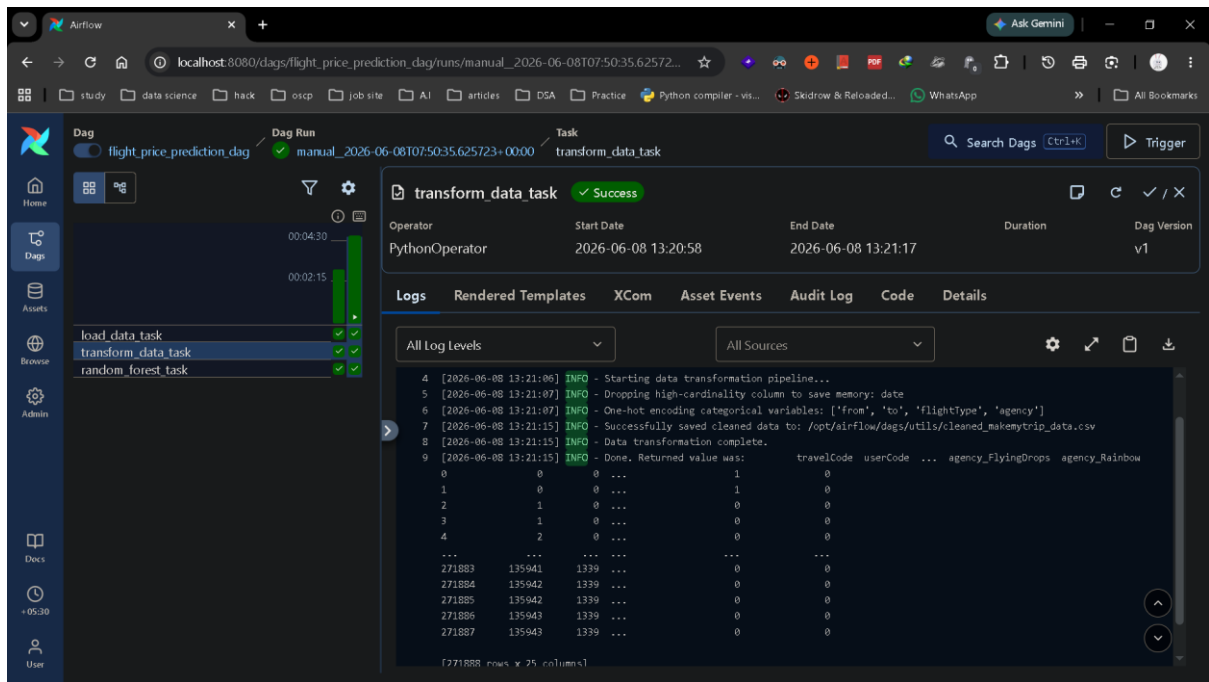
Task Run Summary:

Operator	Start Date	End Date	Duration	Dag Version
PythonOperator	2026-06-08 13:20:43	2026-06-08 13:20:50	00:00:14.796	v1

Logs:

```
[2026-06-08 13:20:47] INFO - Starting data ingestion from: /opt/airflow/data/flights.csv
[2026-06-08 13:20:48] INFO - Successfully ingested 271888 rows.
[2026-06-08 13:20:48] INFO - Done. Returned value was:
  travelCode  userCode  ...  agency  date
0            0        0 ... FlyingDrops 09/26/2019
1            0        0 ... FlyingDrops 09/30/2019
2            1        0 ... CloudFy    10/03/2019
3            1        0 ... CloudFy    10/04/2019
4            2        0 ... CloudFy    10/10/2019
...          ...      ...      ...
271883       135941    1339 ... CloudFy    07/12/2020
271884       135942    1339 ... CloudFy    07/16/2020
271885       135942    1339 ... CloudFy    07/20/2020
271886       135943    1339 ... CloudFy    07/23/2020
271887       135943    1339 ... CloudFy    07/26/2020

[271888 rows x 10 columns]
```

- **Caption:** Apache Airflow Graph View displaying successful, green-lit execution workflows for automated travel analytics.

6. CI/CD PIPELINE WITH JENKINS

- **Objective:** Established a Continuous Integration and Continuous Deployment (CI/CD) pipeline inside Jenkins. Code updates triggered via repository changes automatically run tests, rebuild Docker images, and deploy updates with zero downtime.

- **System View:**

The screenshot shows the Jenkins job view for 'flight_price_prediction'. The job status is 'flight_price_prediction' with a green checkmark indicating success. The left sidebar contains navigation options: Status, Changes, Workspace, Build Now, Configure, Delete Project, GitHub Hook Log, and Rename. The main area displays 'Permalinks' for various build types: Last build (#14), Last stable build (#14), Last successful build (#14), Last failed build (#13), Last unsuccessful build (#13), and Last completed build (#14). Below this, a 'Builds' section shows a list of recent builds with their status (green checkmark for success, red X for failure) and timestamps.

Build #	Status	Timestamp
#14	Success	8:10pm
#13	Failure	8:03pm
#12	Success	7:59pm
#11	Failure	7:53pm

The screenshot shows the Jenkins dashboard. The left sidebar includes 'New Item', 'Build History', 'Build Queue', and 'Build Executor Status'. The main area displays a table of build history for the 'flight_price_prediction' job. The table has columns for Status (S), Workflow (W), Name, Last Success, Last Failure, and Last Duration. The current build is shown as successful with a green checkmark.

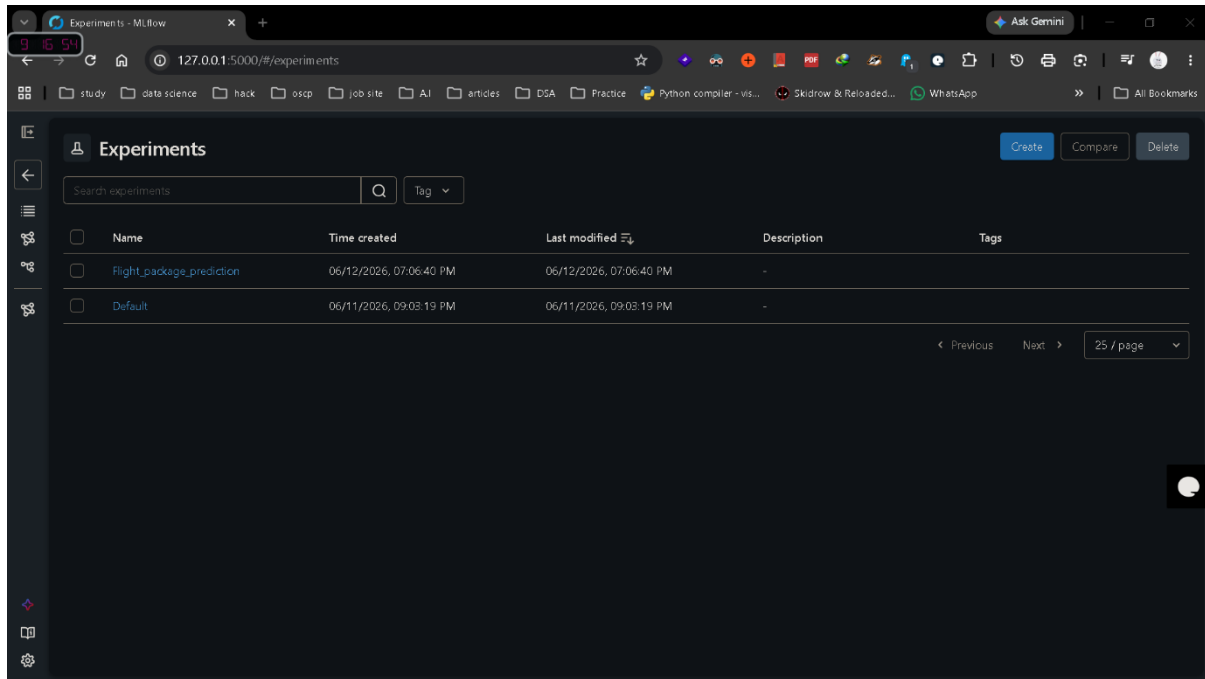
S	W	Name	Last Success	Last Failure	Last Duration
✓	☁	flight_price_prediction	3 min 3 sec #14	10 min #13	2 min 13 sec

At the bottom right, there is a 'Summarize This Page' button and a 'REST API' link.

- **Caption:** Jenkins automation pipeline showcasing a flawless automated build, verification, and deployment execution cycle.

7. MODEL TRACKING WITH MLFLOW

- **Objective:** Integrated MLflow for enterprise-grade experiment tracking and versioning. The setup tracks running hyperparameters, performance logs, evaluation charts, and records production-ready models within the centralized model registry.
- **System View:**



Flight_package_prediction > Runs >

big-stoat-578

OverviewModel metricsSystem metricsTracesArtifacts

Description

No description

Metrics (5)

Search metrics

Metric	Value	Models
CV_R2_Mean	0.9948848607450207	random_forest_model
MSE_Test	667.9228544310884	random_forest_model
RMSE_Test	25.844203497710822	random_forest_model
MAE_Test	13.807689607488152	random_forest_model
R2_Test	0.9949311156521514	random_forest_model

Parameters (5)

Search parameters

Parameter	Value
test_size	0.2
random_state	42
cv_folds	5
max_depth	10
n_estimators	100

Logged models (1)

Model attributes							No data
Type	Step	Model name	Status	Created	Registered models	Dataset	CV_R2_M

About this run

Created at

06/12/2026, 07:06:43 PM

Created by

sand

Experiment ID

1

Status

Finished

Run ID

9c097bfa2af94070916ea8956172a659

Duration

10.7min

Source

flight_price_pred_mflflowipyrb

Registered prompts

—

Datasets

None

Tags

Add tags

Registered models

FlightPackagePriceModel

v2

Flight_package_prediction > Runs >

big-stoat-578

OverviewModel metricsSystem metricsTracesArtifacts

Parameters (5)

Search parameters

Parameter	Value
test_size	0.2
random_state	42
cv_folds	5
max_depth	10
n_estimators	100

Logged models (1)

Model attributes							No data
Type	Step	Model name	Status	Created	Registered models	Dataset	CV_R2_M
Output	0	random_forest_m	Ready	2 hours ago	FlightPackagePriceModel	-	0.994884

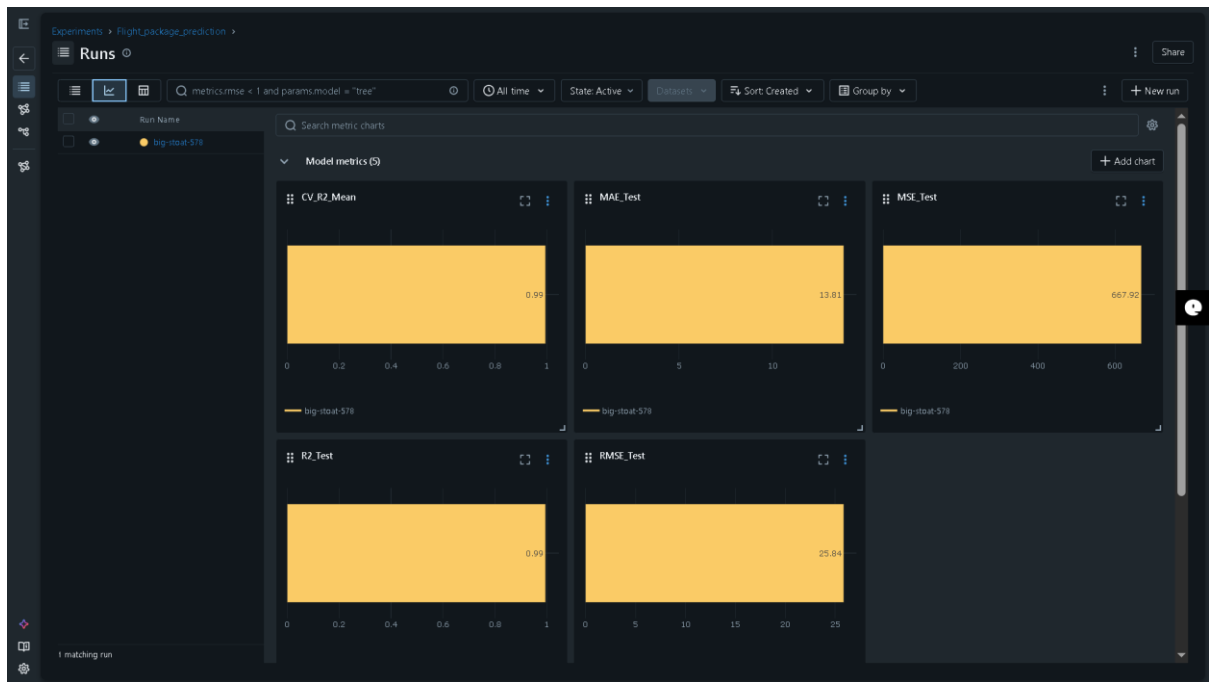
Tags

Add tags

Registered models

FlightPackagePriceModel

v2



Runs - Experiment 1 - MLflow

Ask Gemini

127.0.0.1:5000/#/experiments/1/runs?searchFilter=&orderByKey=attributes...

study | data science | hack | osp | job site | AI | articles | DGA | Practice | Python compiler - vs... | Skidrow & Reloaded... | WhatsApp

Experiments > Flight_package_prediction >

Runs

metrics.rmse < 1 and params.model = "tree" | All time | State: Active | Datasets | Sort: Created | Columns | Group by | + New run

Run Name	Created	Dataset	Duration	Source	Model
puzzled-penguin-224	11 minutes ago	-	11.0min	flight_pri...	-
big-stout-578	2 hours ago	-	10.7min	flight_pri...	FlightPackagePriceModel ...

2 matching runs

Run 8a852ddf50e6449d9df32eeab5520fdb

127.0.0.1:5000/#/experiments/1/runs/8a852ddf50e6449d9df32eeab5520fdb

Flight_package_prediction > Runs > puzzled-penguin-224

Overview Model metrics System metrics Traces Artifacts

Search metrics

Metric	Value	Models
CV_R2_Mean	0.9948848607450207	random_forest_model
MSE_Test	667.9228544310884	random_forest_model
RMSE_Test	25.844203497710802	random_forest_model
MAE_Test	13.807489607488152	random_forest_model
R2_Test	0.9949311156521514	random_forest_model

Parameters (5)

Search parameters

Parameter	Value
test_size	0.2
random_state	42
cv_folds	5
max_depth	10
n_estimators	100

Logged models (1)

Model metrics

Status: Finished

Run ID: 8a852ddf50e6449d9df32eeab5520fdb

Duration: 11.0min

Source: flight_price_pred_mflflowipyrb

Registered prompts: —

Datasets: None

Tags: Add tags

Registered models: FlightPackagePriceModel v3

Run 8a852ddf50e6449d9df32eeab5520fdb

127.0.0.1:5000/#/experiments/1/runs/8a852ddf50e6449d9df32eeab5520fdb

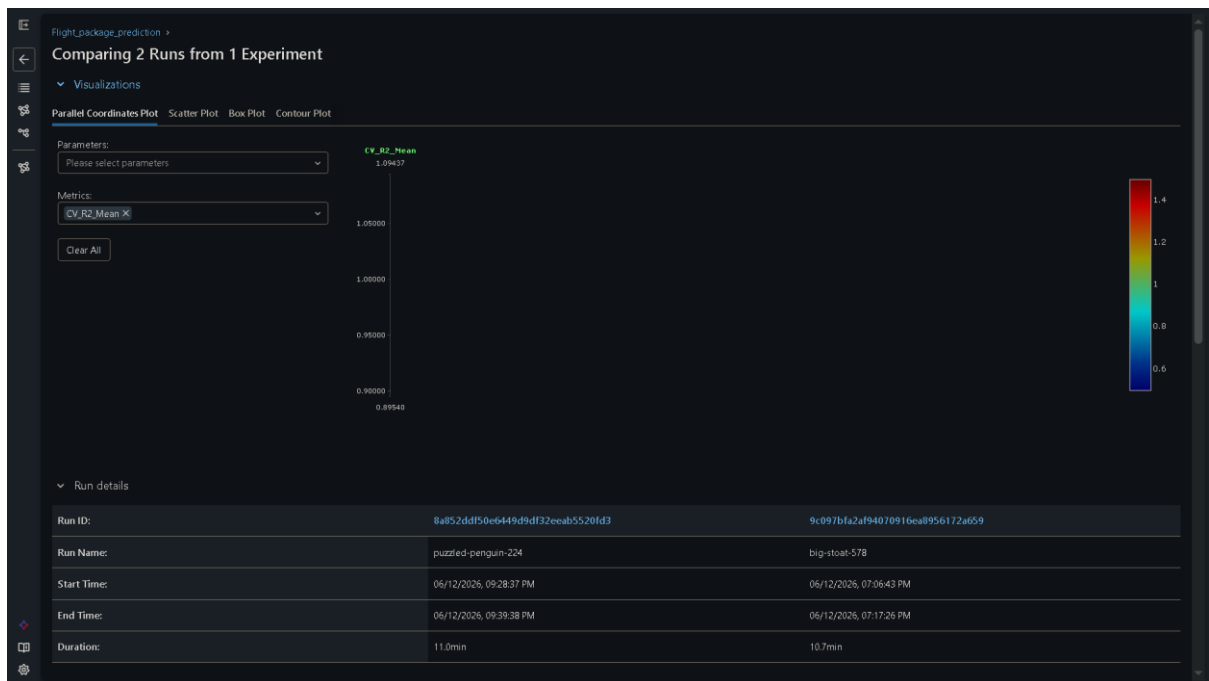
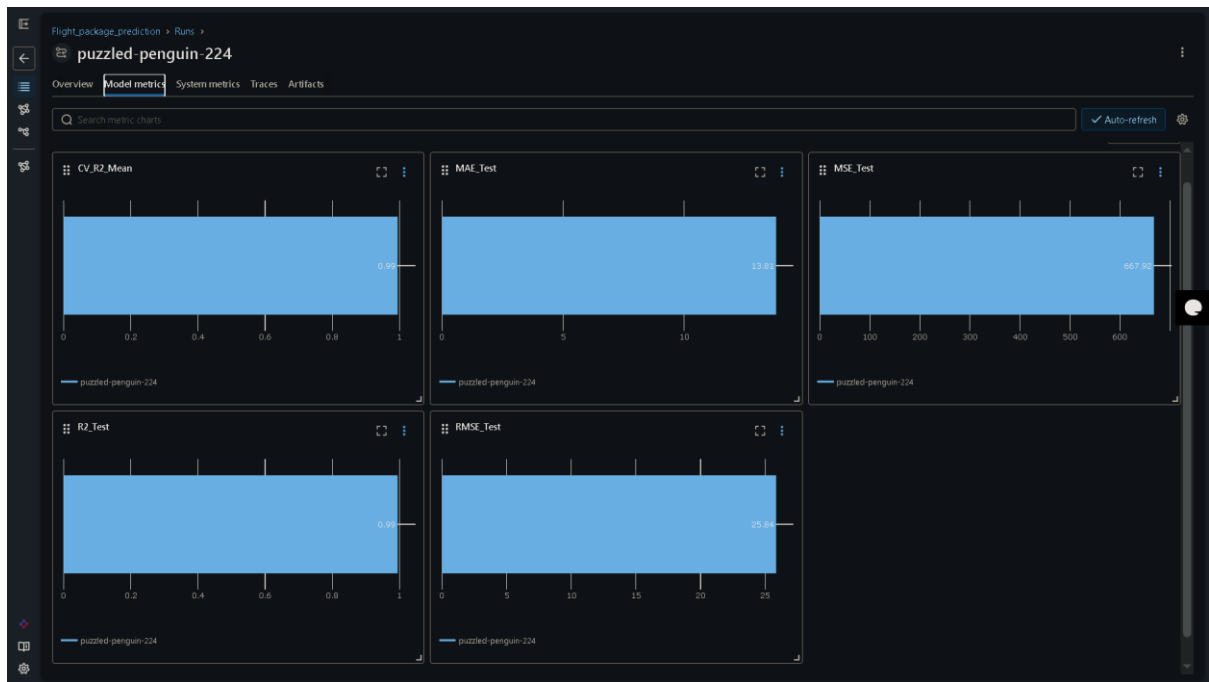
Flight_package_prediction > Runs > puzzled-penguin-224

Overview Model metrics System metrics Traces Artifacts

Logged models (1)

Model attributes

Type	Step	Model name	Status	Created	Registered models	Dataset	No data
Output	0	random_forest_mc	Ready	1 minute ago	FlightPackagePriceModel v3		CV_R2_Mean: 0.994884



Parameters		
☐ Show diff only		
cv_folds	5	5
max_depth	10	10
n_estimators	100	100
random_state	42	42
test_size	0.2	0.2
Metrics		
☐ Show diff only		
CV_R2_Mean	0.995	0.995
MAE_Test	13.81	13.81
MSE_Test	667.9	667.9
R2_Test	0.995	0.995
RMSE_Test	25.84	25.84
Artifacts		
No common artifacts to display.		
Tags		
☐ Show diff only		
No tags to display.		

8. GENDER CLASSIFICATION MODEL

- Objective:** Deployed a demographic classification framework designed to categorize a user's gender from booking data vectors, helping drive specialized user experience loops and targeted travel marketing layers.
- System View:**

Gender Classification Model

127.0.0.1:8000

Gender Classification Model

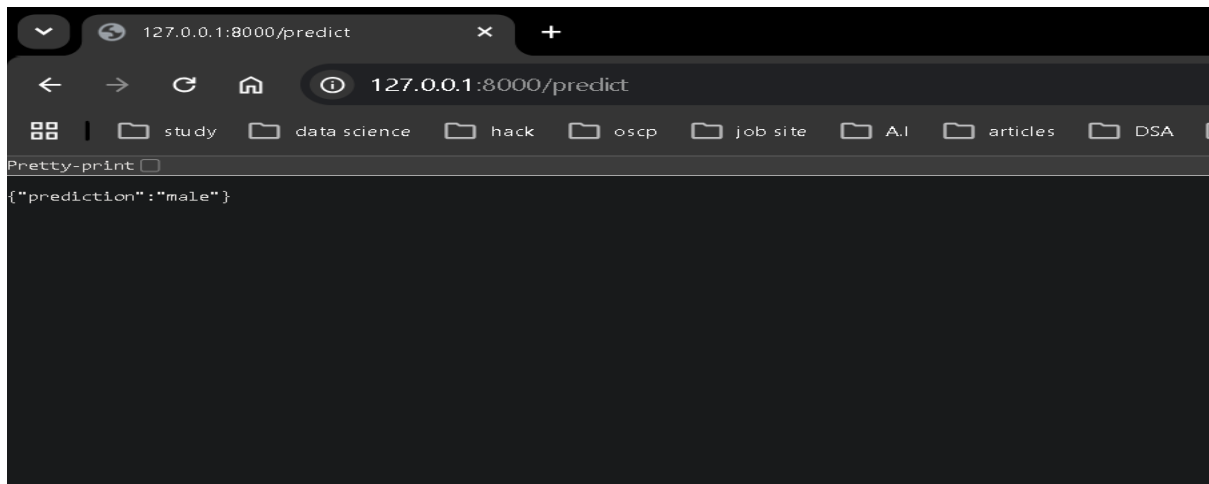
Username:
Roy Braun

Usercode:
55

Traveller_Age:
54

Company name:
Acme Factory

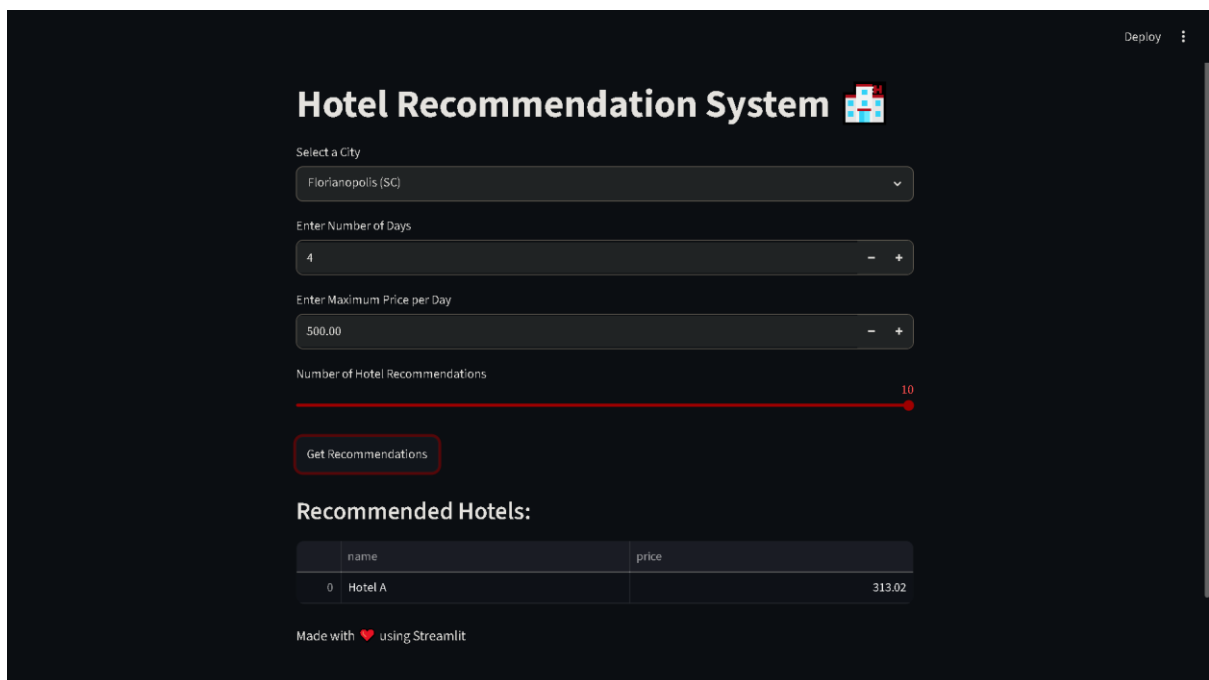
Predict



Caption: Functional model validation and layout mapping for the demographic Gender Classification component.

9. TRAVEL RECOMMENDATION MODEL (HOTEL RECOMMENDER STREAMLIT APP)

- **Objective:** Programmed a custom recommendation system running Matrix Factorization/Collaborative Filtering on historical booking datasets to provide optimized hotel lodging options. Deployed as an interactive Streamlit web application providing clear insights and visualization dashboards.
- **System View:**



Caption: Live Streamlit web application dashboard displaying personalized hotel suggestions and travel insights.